Characteristics of Combustion and Pollutant Formation in Swirling Flames

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Detailed measurements are presented of gas species concentrations and temperature in the flame formed around a fuel (propane) jet surrounded by a swirling air flow. Attention is given to the interrelation between the flame structure and the formation and emission characteristics of air pollutants such as NO_x, CO, and hydrocarbons.

It is noted that the mixing and combustion of the fuel layer flowing around the recirculation zone in the swirling flame structure control the formation and emission of NO_x, CO, and hydrocarbons.

Two typical flame configurations are found to arise according to the primary air ratio:

Low NO_x formation occurs in a flame (type B) having a high primary air ratio. This results from the fact that the fuel layer around the recirculation zone burns at relatively low temperature because of dilution by mixing with the surrounding excess air prior to the combustion, and the temperature of the burnt hydrocarbons are liable to be exhausted because the hydrocarbons in the fuel layer are excessively diluted and spill away through the circumferential part of the bulk flow.

For another flame (type A) having a low primary air ratio, increased NO_x is formed in the flame and appears in the exhaust because the fuel layer burns at relatively high temperature and the slow mixing retards the temperature decay.

Effects of a wide range of combustion parameters on NO_x, CO, and hydrocarbon emissions are examined in connection with the flame configuration.

1. INTRODUCTION

Flames with swirl are often used in practical combustion systems because the swirl-induced recirculation zone stabilizes the flame.

Numerous studies have been concerned with the swirling flame as reviewed in Refs. [1] and [2]. They are concerned with the flow without flame [3–5] and with flame [6–10], flame configuration [11–13], flame stability [13,14], and noise emission [14, 15].

As for the formation and emissions of air pollutants such as NO_x, CO, and hydrocarbons in swirling flames, studies are limited [16–21] and the swirl has inconsistant effects of increasing [14, 18] or decreasing [19, 20] NO_x emission probably because of different flame structure. Little attention has been paid to the processes of the emission of CO and unburnt hydrocarbons in relation to the flame structure. Detailed measurements of gas species and temperature in swirling flames seem to be necessary to reveal the interrelation between the flame structure and the formation and emission characteristics.

In the present study, detailed measurements of gas species concentration and temperature were made in two different types of flames formed around a fuel jet surrounded by swirling air flow, and effects of a wide range of combustion parameters on pollutant emissions were examined in order to interpret the characteristics and processes of pollutant formation and emissions as they relate to the flame structure.

2. EXPERIMENTAL APPARATUS AND PROCEDURE

The combustor arrangement and the detail near the fuel nozzle tip are shown in Fig. 1. Gaseous fuel

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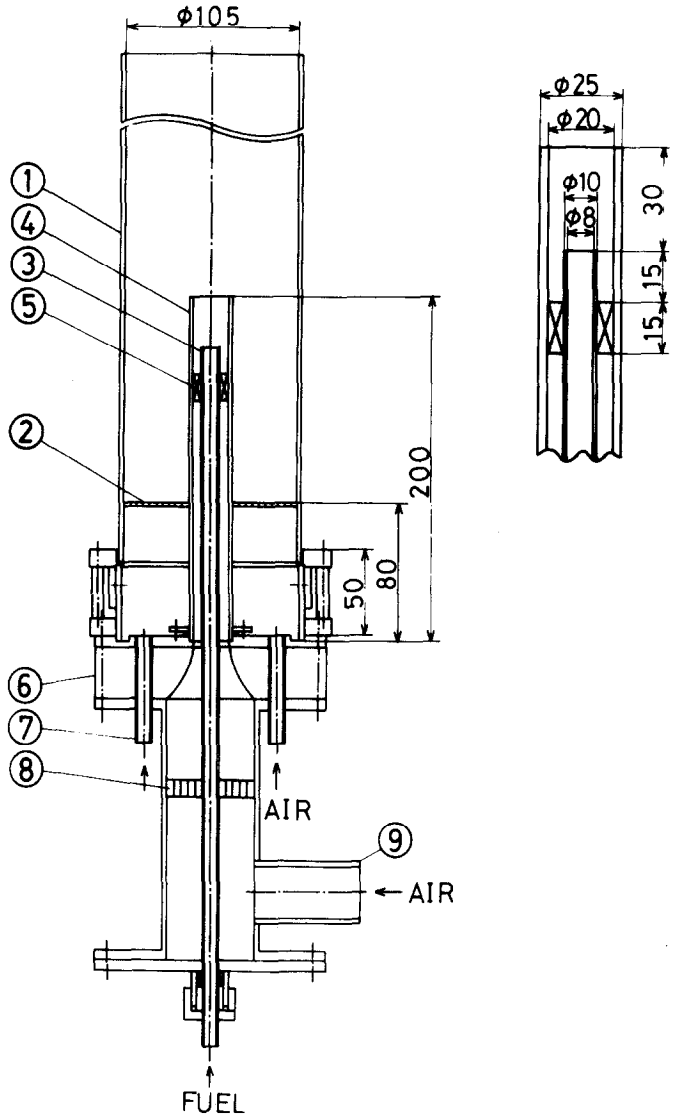

Fig. 1. Combustor arrangement.

(propane) is supplied through a fuel nozzle ③ surrounded by swirling primary air which flows through a throttle ⑥, a swirler composed of eight 45°-inclined vanes, 0.2 mm in thickness ⑤, and a Pyrex glass straight air nozzle ④. Secondary air is introduced through 16 tubes arranged in a circle at regular intervals ⑦ and a multi-hole plate ②. The swirling turbulent diffusion flame is formed at atmospheric pressure in a Pyrex glass tube ①.

The principal experimental parameters are fuel flow rate, primary air ratio λ_1, and secondary air ratio λ_2. The primary or secondary air ratio is defined by primary or secondary air flow rate divided by stoichiometric air flow rate, respecively. Total air ratio λ_T is the sum of the primary and secondary air ratio.

Detailed profiles of concentrations of gas species (H_2, O_2, N_2, CO_2, CO, NO, NO_x, and hydrocarbons) and temperature in typical flames and exhausted gas species concentrations are measured. The gas in the flame is sampled by a water-cooled stainless steel probe, 0.7 mm in inner diameter. The

probe is made as thin as possible to be 2.76 mm in outer diamter, including the water jacket, in order to minimize the disturbance of the flow, especially in the recirculation zone. The probe is cooled by preheated water and the sampling line of Teflon tubing is covered with an electric heater to prevent water condensation in the line. NO and NO_x are detected by a chemiluminescence detector equipped with an NO_2–NO converter. Other gas species noted above are analyzed by a gas chromatograph. Gas temperature is measured by a thermocouple (Pt.-Pt. Rh. 13% or Pt. Rh. 6%-Pt. Rh. 30%, 0.1 mm in diameter) coated with MgO. Radiation loss is not considered.

3. RESULTS AND DISCUSSIONS

3.1. Typical Flame Configuration

Two typical types of nonluminous flame arise when primary and secondary air ratio are changed keeping the fuel and total air flow rate constant. Photographs and sketches of the flame types A and B are shown in Fig. 2. Type B is formed for higher primary air ratio than type A. The flame conditions are noted in the figure caption. The swirling primary air serves to form a recirculation zone which is visualized by the light emission upon adding a small amount of NaCl to the flow, and the position and the size of the recirculation zone are observed. The recirculation zone of the reversed cone shape is drawn back into the primary air nozzle and extends downstream to a location 25 or 35–40 mm from the primary air nozzle tip for the flame of type A or B, respectively. The recirculation zone is filled with high-temperature burnt gas to anchor the flame and the fuel flows mainly around the recirculation zone forming a fuel layer as shown schematically in the sketches of Fig. 2. This is confirmed by the measurements of gas species concentration and temperature in the next section. In the flame of type A, the fuel layer burns to form a ring flame around the recirculation zone that causes the sudden spread of the flame width. In the flame of type B, however, the ring flame is not observed because the fuel layer around the recirculation zone burns at the downstream section after dilution by the excess air.

The mode of the burning of the fuel layer around the recirculation zone characterizes the typical flame configuration and structure.

3.2. Profiles of Gas Species Concentration and Temperature in Typical Flames

Figures 3–8 show the profiles of gas concentrations and temperature T in a typical flame of type A or B whose flame conditions are the same with those listed in the caption of Fig. 2. The abscissa R is the radial distance from the flame axis and X is the axial distance from the primary air nozzle tip. The concentrations are plotted on a wet basis. HC in the figure means the total hydrocarbon concentration reduced in terms of carbon number of 3. ϕ is the local equivalence ratio calculated from the measured gas concentrations.

The following are noted from Figs. 3–5 for the flame of type A.

1. The cross section at $X = 15$ mm of Figs. 3a and 4 crosses the recirculation zone. In the recirculation zone, high concentrations of the burnt gas species CO_2 and H_2O are observed and temperature is high. The HC concentration profile has a saddle shape and the peak of the HC concentration is located at the circumferential part of the recirculation zone where ϕ also has a peak value. C_3H_8 provides the main part in HC concentration. The above observations indicate that the fuel (C_3H_8) flows mainly along the circumferential part of the recirculation zone to form the fuel layer around it. The fuel at the outer part of the fuel layer is not yet burnt even though the mixing of the fuel and air proceeds.
2. The section at $X = 30$ mm of Fig. 3b and 4 crosses the ring flame observed in Fig. 2a. The HC concentration profile has a saddle shape and the fuel layer is still observed. The concentrations of the burnt gases (H_2O, CO_2) and the temperature all increase at the outer part of the fuel layer which corresponds to the formation of the ring flame around the recirculation zone.
3. The section at $X = 50$ mm of Figs. 3c and 4 crosses the end of the ring flame. The HC concentration becomes much lower than that of the section at $X = 15$ or 30 mm, which indicates the fuel layer burns mostly at the ring flame. The equiva-

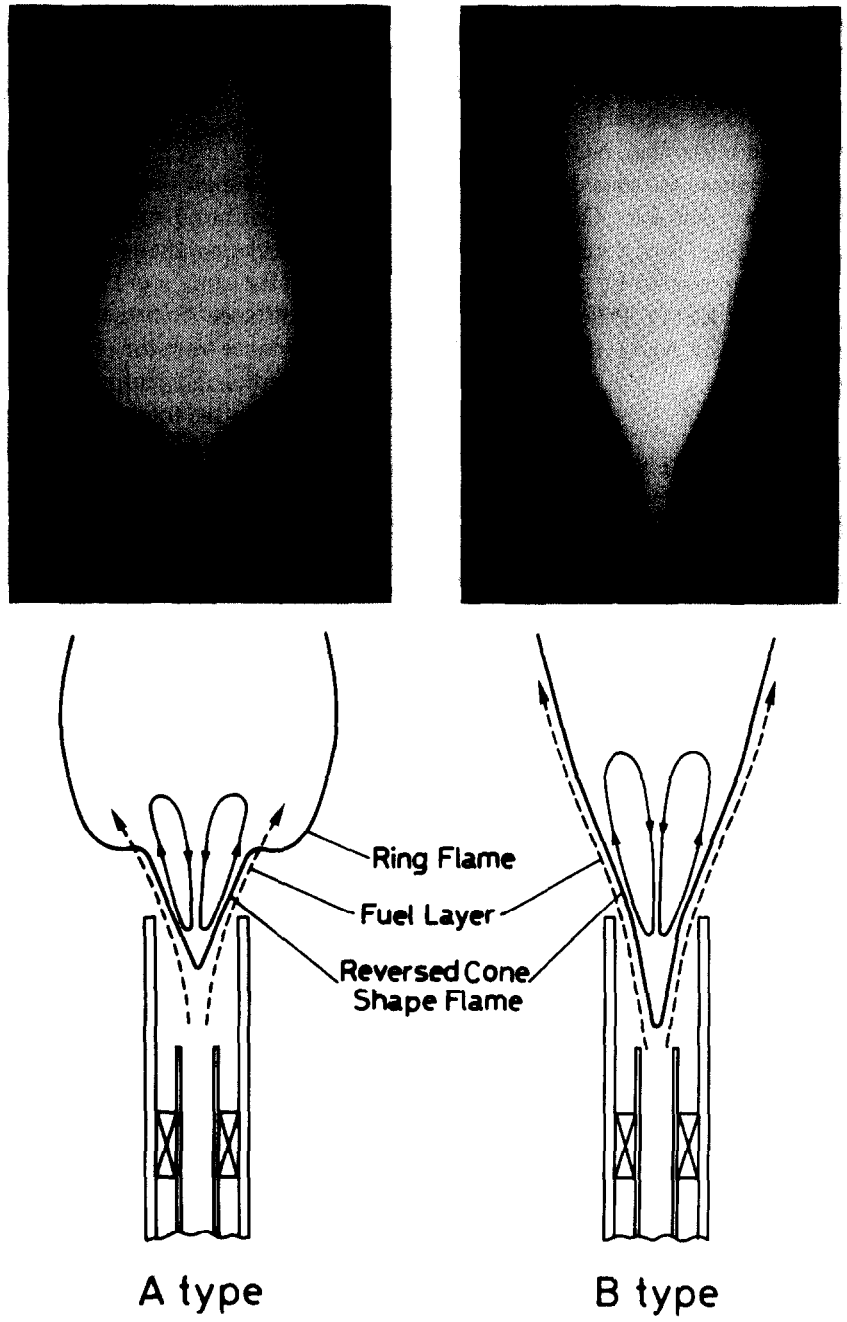

Fig. 2. Photographs and sketches of the typical flames of (a) type A and (b) type B. Fuel flow rate = 10 Nl/min, λ_T = 2.0: type A, λ_1 = 0.6 and λ_2 = 1.4; type B, λ_1 = 1.2 and λ_2 = 0.8.

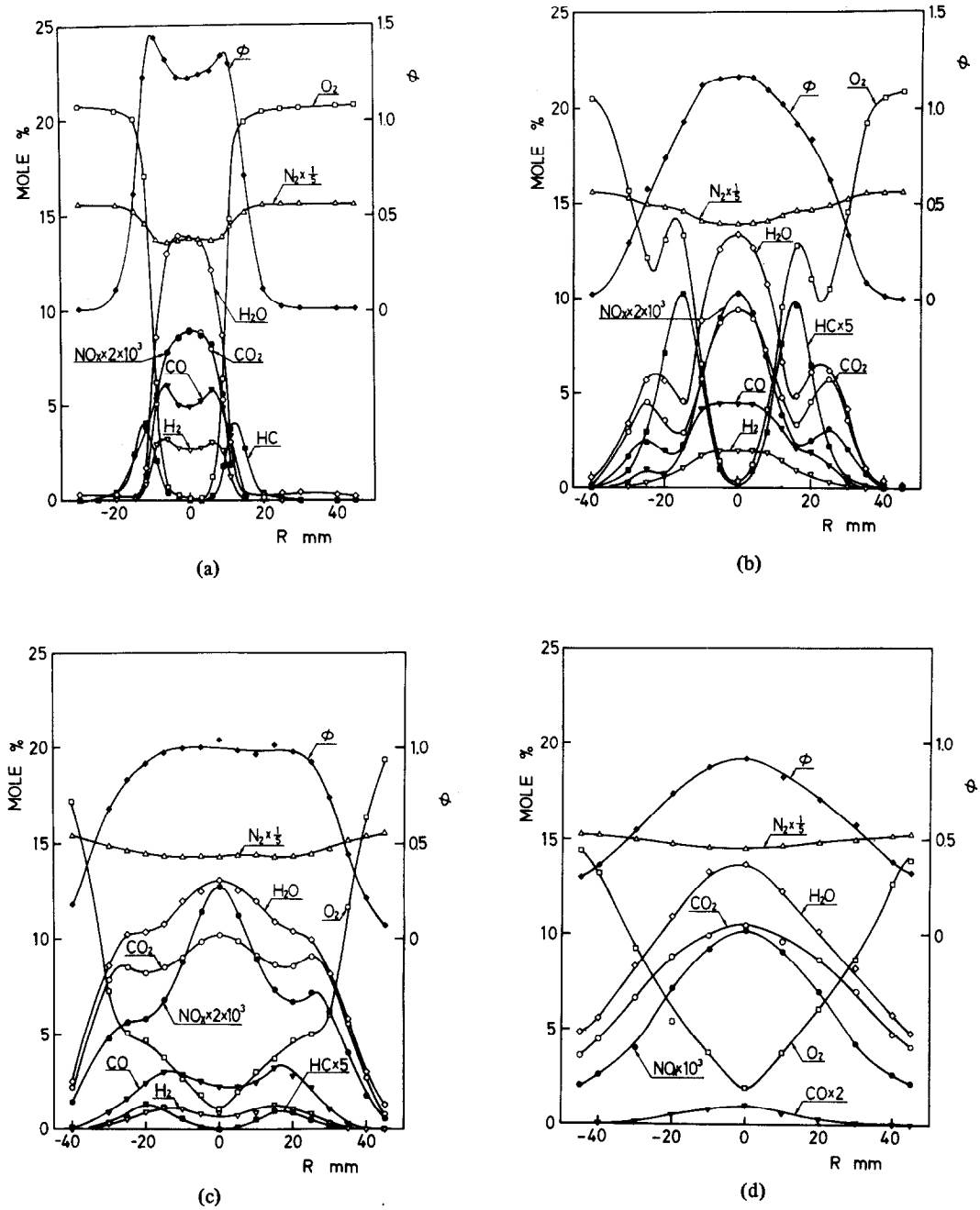

Fig. 3. Radial profiles of species concentrations for the flame of type A: (a) X = 15 mm; (b) X = 30 mm; (c) X = 50 mm; (d) X = 150.

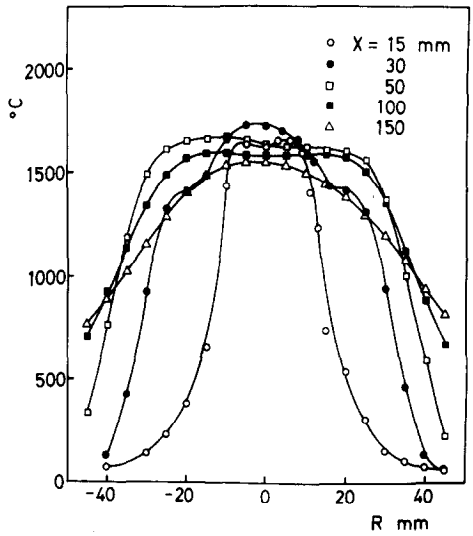

Fig. 4. Radial profiles of temperature for the flame of type A.

lence ratio in the ring flame is nearly stoichiometric.
4. The temperature remains relatively high in the flame of type A as in Fig. 4, and consequently the NO_x concentration continues to increase along with the flow even at $X = 50$–150 mm as shown in Fig. 5. The slow temperature decay results from the retardation of mixing with the surrounding air

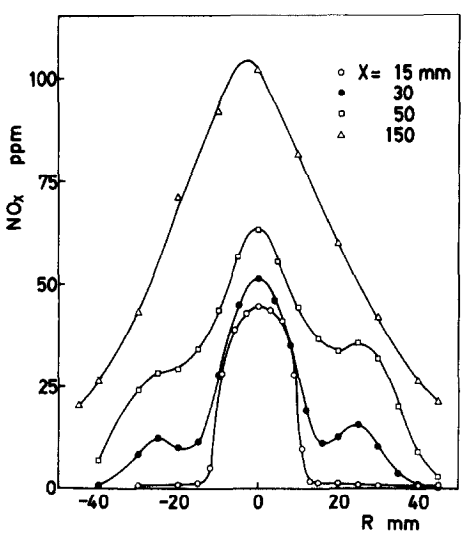

Fig. 5. Radial profilles of NO_x concentration for the flame of type A.

because the equivalence ratio ϕ remains high in the downstream at $X = 150$ mm in Fig. 3d.

The following are noted from Figs. 6–8 for the flame of type B.

1. The sections at $X = 15$ and 30 mm of Figs. 6a and 6b cross the recirculation zone located at the central part where the concentrations of burnt gas species (H_2O and CO_2) are high. The fuel layer is recognized at the circumferential part around the recirculation zone. The fuel layer is rapidly premixed with the surrounding air without forming the flame at the outer part of the fuel layer, and the equivalence ratio at the center of the fuel layer reaches the lean value of $\phi = 0.6$ at $X = 30$ mm.
2. The fuel layer burns in the downstream region of $X = 50$ mm to form a weak reaction zone where the equivalence ratio is 0.4–0.5. However, unburnt hydrocarbons are liable to survive in the circumferential part of the flow even at $X = 100$ mm of Fig. 6d, which results from the temperature decay accompanied by the rapid mixing with the excess air as shown in Fig. 7.
3. The profiles of NO_x concentration in the flame of type B as shown in Fig. 8 indicate that the relatively high NO_x concentration observed is limited to the recirculation zone where temperature is also high. The rate of NO_x formation is presumed to be very low in the downstream where NO_x concentration decays because of the diffusion effect on the center line. The low NO_x formation in the downstream comes from the rapid temperature decay caused by rapid mixing with the surrounding excess air which consequently results in low NO_x emission.

The characteristics of NO_x formation and emission in the present flame of type A and B are supposed to be mainly dominated by the NO_x formation in the postreaction zone in the downstream through the Zeldovich mechanism even though some amount of NO_x should be formed by other mechanisms related to HCN species as in turbulent jet diffusion flames [9].

3.3. Characteristics of NO_x, HC, and CO Emissions

Effects of various combustion parameters and flame configurations on NO_x, HC, and CO concentrations exhausted from the exit of the combustor are examined.

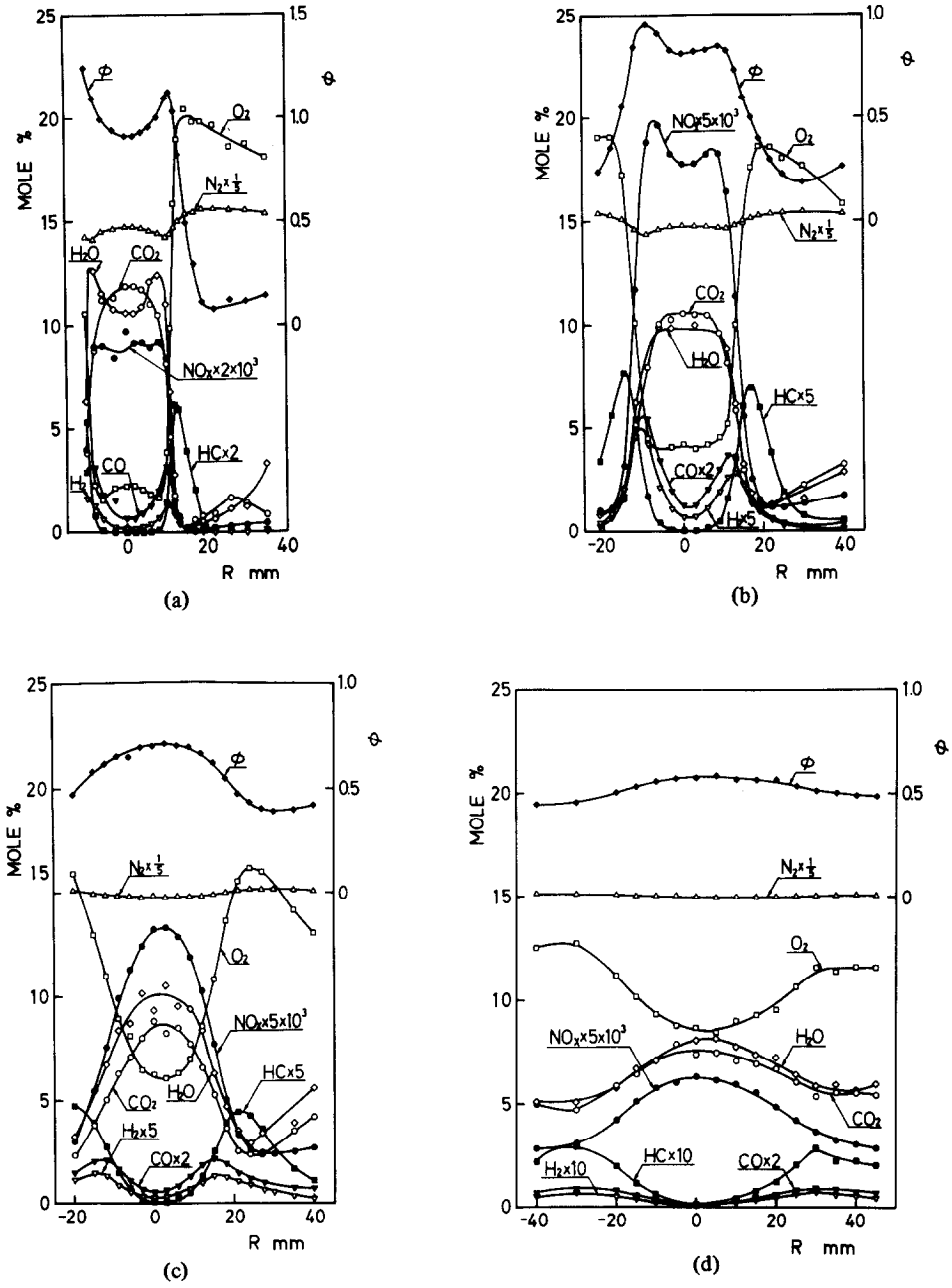

Fig. 6. Radial profiles of species concentrations for the flame of type B: (a) X = 15 mm; (b) X = 30 mm; (c) X = 50 mm; (d) X = 100 mm.

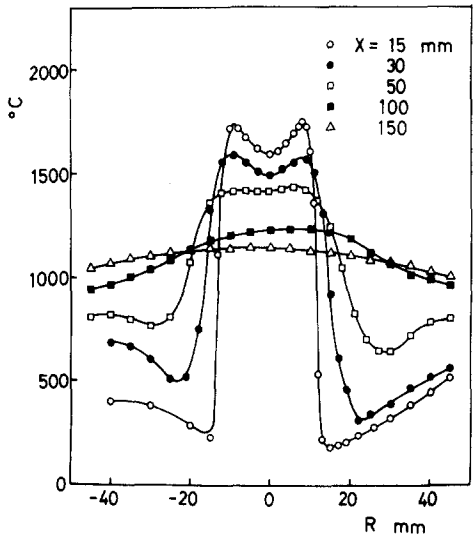

Fig. 7. Radial profiles of temperature for the flame of type B.

In Fig. 9, exhausted NO_x concentrations versus primary air ratio λ_1 are shown taking total air ratio $\lambda_T (=\lambda_1+\lambda_2)$ and fuel flow rate as the parameters. NO_x* means the NO_x concentration corrected for dilution by excess air and is proportional to the amount of NO_x exhausted per unit mass of fuel. *

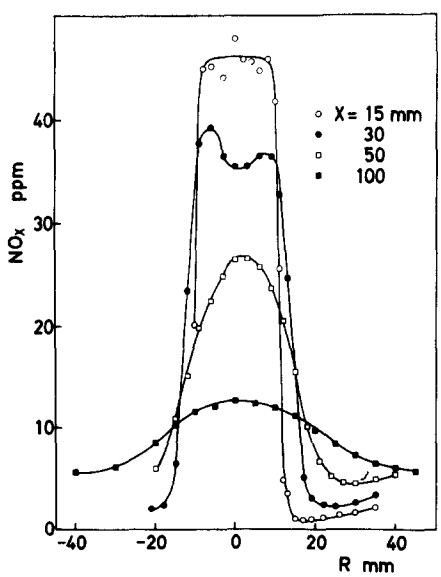

Fig. 8. Radial profiles of NO_x concentration for the flame of type B.

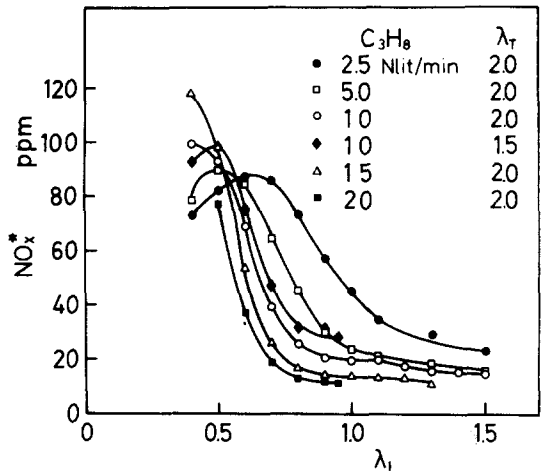

Fig. 9. NO_x emission versus primary air ratio.

labeled to other species has the same meaning. The following are noted from Fig. 9.

For a primary air ratio larger than 0.5, a larger primary air ratio results in lower NO_x* emission, which reaches the very low value of 10–25 ppm. The tendency of NO_x* to decrease for larger λ_1 tends to be saturated for a larger primary air ratio. A larger fuel flow rate results in lower NO_x* emission when keeping primary and total air ratio constant. Flame configurations of high or low NO_x* emission tend to correspond to flames of type A or B, respectively. Concentrations of exhausted HC and CO are too low to detect for the flame in the case of Fig. 9 ($\lambda_T = 1.5$ or 2.0).

In Fig. 10, exhausted NO_x, NO, HC, and CO versus total air ratio λ_T are shown keeping fuel flow rate and primary air ratio λ_1 constant. The flames of Fig. 10 are of type B regardless of λ_T, and consequently NO_x* lies below 20 ppm. It is noted, however, that HC and CO emissions increase significantly for higher total air ratio than $\lambda_T = 2.0$, which is caused by the fact that the fuel layer around the recirculation zone in the flame of type B is not easily oxidized in the downstream because the temperature decays by rapid mixing with excess air even though the flame in the recirculation zone is very stable.

NO_2 ($=NO_x - NO$) represents a large fraction of the total NO_x for flames of high total air ratio as shown in Fig. 10.

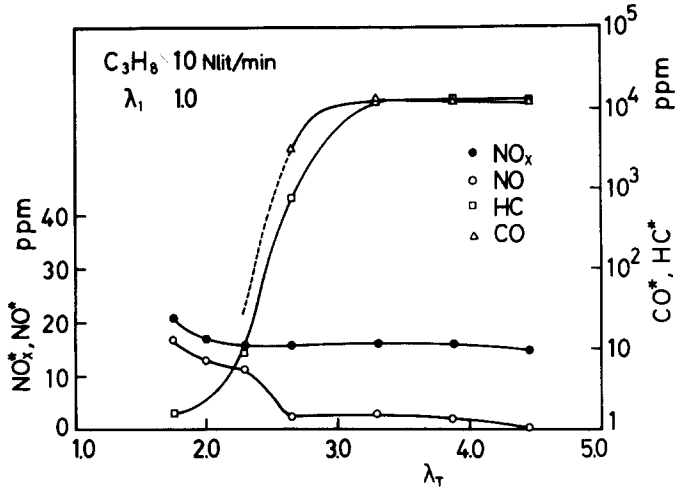

Fig. 10. NO_x, NO, CO, and HC emissions versus total air ratio.

Two types of probes were used to examine the effects of the NO–NO_2 reaction in the probe. One is 0.7 mm in inner diameter, with water cooling, and the other is 5 mm in inner diameter, without cooling. The sampled gas passing the each probe should experience a very different temperature history. When the exhausted gas at the combustor exit was probed, the two types of probes made no difference in the measured concentrations of NO_x and NO. Thus, it seems probable that NO_2 ($=NO_x-NO$) is really exhausted at the exit as derived from Fig. 10. However, when the sampling was made within the flame, the two types of probes yielded the same NO_x, but a significantly different NO concentration, which indicates significant conversion from NO to NO_2 or vice versa in the probe. Hence the detection of NO_2 concentration in the flame is ambiguous even though NO_2 formation via NO is highly probable in the combustor [21, 23, 24].

In Fig. 11, equiconcentration curves of exhausted NO_x are shown by solid lines, where primary air ratio λ_1 and fuel flow rate are taken as the coordinates keeping total air ratio λ_T constant at 2.

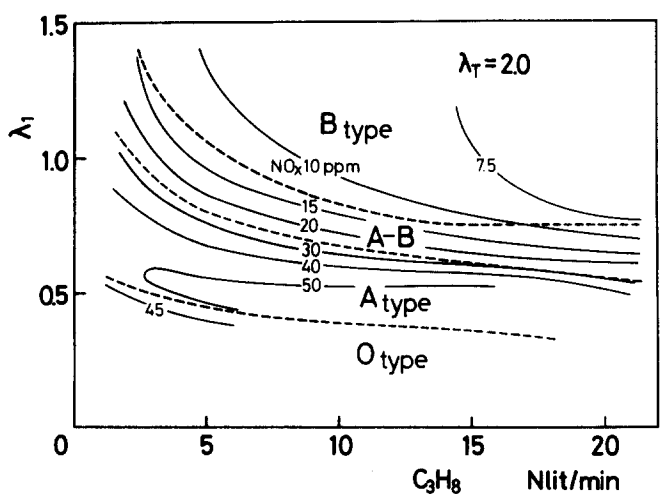

Fig. 11. Equiconcentration curves of NO_x emission and flame type map versus primary air ratio and fuel flow rate.

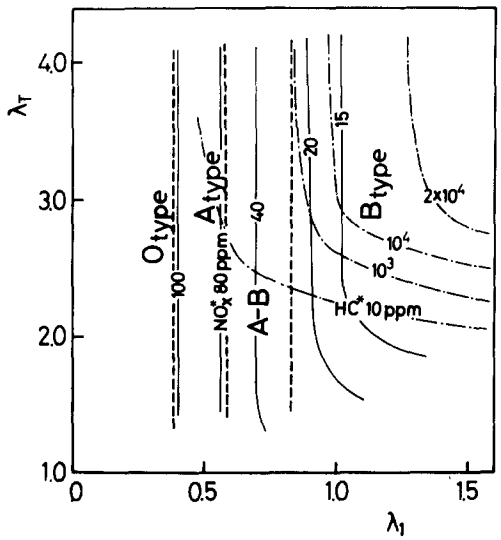

Fig. 12. Equiconcentration curves of NO_x and HC emissions and flame type map versus primary and total air ratio. (Fuel flow rate is 10 Nl/min).

Broken lines classify the flame types. A and B types are the typical flame configurations as described above. A–B type is the transition type between A and B types. The flame of O type is a long flame having no distinctive recirculation zone. The O type arises for primary air ratio less than 0.5.

The equiconcentration curves are nearly parallel to the boundaries of the flame types, which indicates that the NO_x emission is closely connected with the flame configuration. Low NO_x emission can be reached in the flame of type B.

In Fig. 12, equiconcentration curves of exhausted NO_x and HC are shown by solid and chain lines, respectively, where total and primary air ratio are taken as the coordinates, keeping fuel flow rate constant at 10 Nl/min. Broken lines are the boundaries classifying flame types. NO_x emission and the flame type are mainly dominated by the primary air ratio. HC emission is affected by both primary and total air ratio. High primary and total air ratios result in significant HC emission. When the total air ratio is less than 2.0, the HC emission is low even in the flame of B type of high primary air ratio. This results from the fact that the temperature after the mixing of the secondary air in the downstream is high enough to oxidize hydrocarbons.

4. CONCLUSIONS

1. The structure of the flame formed around a fuel jet surrounded by swirling air is characterized by the mixing and combustion of the fuel layer flowing around the high-temperature recirculation zone. Two typical flame configurations (types A and B) arise according to the primary air ratio. In the flame of type A, the fuel layer burns near stoichiometric condition to form a ring flame around the recirculation zone. In the flame of type B, the fuel layer is excessively diluted by the mixing with the surrounding air, and it burns in the downstream without forming a distinctive ring flame. The flame of type A occurs for lower primary air ratio than type B.
2. NO_x formation and emission are significantly affected by the flame structure. Low NO_x emission in the flame of type B occurs because the fuel layer burns at relatively low temperature as the result of dilution by excess air prior to the combustion, while the temperature decays rapidly because of the rapid mixing with the surrounding air even though the temperature in the recirculation zone is high. Higher NO_x emission in the flame of type A occurs because high temperature generated in the recirculation zone and the surrounding ring flame continues to the downstream as the result of slow mixing with the excess air and NO_x is increasingly formed in the downstream.
3. Unburnt hydrocarbons are liable to be exhausted in the flame of type B because hydrocarbons in the fuel layer around the recirculation zone are excessively diluted by the surrounding air and spill away through the circumferential part of the bulk flow. However, the hydrocarbons are not exhausted when the total air ratio is less than 2; then the temperature after the mixing of the secondary air in the downstream is high enough to oxidize the unburnt hydrocarbons.

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REFERENCES

1. Lilley, D. G., *AIAA J.* 15:1063 (1977).
2. Syred, N., and Beér, J. M., *Combust. Flame* 23:143 (1974).
3. Chigier, N. A., and Chervinsky, A., *Trans. ASME, J. Appl. Mech.* June:443 (1967).
4. Lilley, D. G., *Acta Astronautica* 3:919 (1976).
5. Mathur, M. L., and Maccallum, N. R. L., *J. Inst. Fuel* May:214 (1967).
6. Chigier, N. A., and Dvorak, K., *Fifteenth Symposium (International) on Combustion,* The Combustion Institute, Pittsburgh, 1975, p. 573.
7. Baker, R. J., Hutchinson, P., Khalil, E. E., and Whitelaw, J. H., *Fifteenth Symposium (International) on Combustion,* The Combustion Institute, Pittsburgh, 1975, p. 553.
8. Beltagui, S. A., and Maccallum, N. R. L., *J. Inst. Fuel* Dec.:183 (1976).
9. Syred, N., Chigier, N. A., and Beér, J. M., *Thirteenth Symposium (International) on Combustion,* The Combustion Institute, Pittsburgh, 1971, p. 617.
10. Syred, N., and Beér, J. M., *Fourteenth Symposium (International) on Combustion,* The Combustion Institute, Pittsburgh, 1973, p. 537.
11. Leuckel, and Fricker, N., *J. of Inst. Fuel* June:103 (1976).
12. Wu, H. L., and Fricker, N., *J. of Inst. Fuel* Sept:144 (1976).
13. Fricker, N., and Lueckel, N., *J. Inst. Fuel* Sept.:152 (1976).
14. Gupta, A. K., Beér, J. M., and Swithenbank, J., *Sixteenth Symposium (International) on Combustion,* The Combustion Institute, Pittsburgh, 1977, p. 79.
15. Gupta, A. K., Syred, N., and Beér, J. M., *Fifteenth Symposium (International) on Combustion,* The Combustion Institute, Pittsburgh, 1975, p. 1376.
16. Sadakata, M., and Beér, J. M., *Sixteenth Symposium (International) on Combustion,* The Combustion Institute, Pittsburgh, 1977, p. 93.
17. Owen, F. K., Spadaccini, L. J., and Bowman, C. T., *Sixteenth Symposium (International) on Combustion,* The Combustion Institute, Pittsburgh, 1977, p. 105.
18. Sadakata, M., Beér, J. M. and Kunii, K., *Kagaku Kogaku Ronbunshu* 4, 5:428 (1978).
19. Tsuji, H., Hori, M., Yasusato, K., and Sonoda, A., *15th Combustion Symposium (in Japan),* 1977, p. 25.
20. Yoshinaga, M., et al., *Tetsu to Koh* 54 (1977).
21. Oven, M. J., Gouldin, F. C., and McLean, W. J., *Seventeenth Symposium (International) on Comustion,* The Combustion Institute, Pittsburgh, 1979, p. 363.
22. Takagi, T., Ogasawara, M., Daizo, M., and Tatsumi, T., *Sixteenth Symposium (International) on Combustion,* The Combustion Institute, Pittsburgh, 1977, p. 181.
23. Cernansky, N. P., and Sawyer, R. F., *Fifteenth Symposium (International) on Combustion,* The Combustion Institute, Pittsburgh, 1975, p. 1039.
24. Takagi, T., Ogasawara, M., and Yoshimoto, T., ASME Paper 78-WA/Fu-2, 1978.

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